

Device-Wise Use Cases

Awesom Living Smart Home Platform

Generated: September 18, 2026 | Covers all development sessions to date

55 Total Use Cases	9 Device Categories	55+ Bugs Fixed
--	---	--------------------------------------

Device	Use Cases
GLK Vital Tracker	12
Motion Sensor	6
Presence Sensor	2
Door/Window Contact Sensors	4
Emergency Button	2
AltumView Sentinare	1
CP Plus Camera	3
Raspberry Pi Hub	14
Cross-Device / Platform	11

1. GLK Vital Tracker (Sleep / Vitals Sensor)

Sole vitals and sleep monitoring device. Replaced Emfit sensor.

- **Sleep date attribution fix** — GLK reports data at 2-6 AM; fixed logic so sessions attribute to the correct sleep date, not the calendar date of the raw timestamp
- **Snoring / activity / restless data leaking from previous day** — data from a prior night was bleeding into the current day's response; fixed fetch window boundaries
- **ROOT CAUSE of empty sleep sessions** — GLK source detection missed cloud webhook format (glk_cloud vs glk_local); sessions appeared empty because the ingestion path didn't recognize the payload
- **previous_sleep_session computation fix** — the 'compared to last night' comparison was pulling the wrong session
- **GLK fetch window fix** — API query window wasn't aligned with actual data arrival pattern (2-6 AM next day)
- **15-minute minimum duration filter** — added to discard spurious micro-sessions (sensor noise, brief power events)
- **Device-offline close in build_sleep_sessions** — sessions left open when device went offline are now properly closed
- **merge_close_sessions widened from 15 min to 30 min** — adjacent sessions separated by brief gaps (bathroom trip) are merged into one continuous night
- **110-minute data gap analysis** — analyzed raw data and proved the gap was the GLK sensor going silent (hardware), not a software bug
- **device_id: null in create_emfit_log** — log creation wasn't attaching the device ID; fixed
- **Backfilled device_id on all 734 existing emfit_logs records** — retroactive data integrity fix
- **get_sleep_date_range window fix** — date range query for sleep data was off by one day

Note: 'Only last sleep session data should come in the APIs' — enforced per user requirement. Resident dropdown kept for GLK pairing only (removed from all other device forms).

2. Motion Sensor (SONOFF SNZB-03 Zigbee)

Room-level motion detection for occupancy and 'no motion' alerts.

- **Motion sensor flickering + stale presence bug fix** — dashboard showed rapid on/off flickering and stale 'motion detected' that never cleared
- **Motion sensor pairing UI** — added 'Paired Motion Sensor' and 'Paired Window/Door Sensor' dropdowns to dashboard device form for door-aware occupancy linking
- **'Too long in room' no-motion alerts — 3 critical bugs fixed:** (1) Alert timer not resetting on new motion events, (2) Grace period calculation wrong, (3) Alert firing during normal occupancy patterns
- **Motion sensor alert grace period bug** — configurable grace period before 'no motion' alert wasn't being respected
- **Motion sensor alerts not firing — 3 root causes:** (1) Zigbee event path not triggering alert evaluation, (2) Home-based device lookup failing, (3) Alert threshold comparison inverted
- **Multiple paired window sensor filter fixes** — filter associating a contact sensor with its paired motion sensor had edge cases where the wrong sensor was matched

3. Presence Sensor (SONOFF SNZB-06P Zigbee)

mmWave-based presence detection — detects stationary persons (sitting/sleeping).

- **Occupancy showing false while person is sitting still** — presence sensor correctly reported occupancy but dashboard showed 'unoccupied' due to logic error in state aggregation
- **Door-aware occupancy logic (complete implementation)** — presence + motion + door sensor data combined: if door opens and presence drops, person left; if presence stays, person is still there despite door activity

4. Door/Window Contact Sensors (SONOFF SNZB-04 Zigbee)

Open/close detection for doors and windows. Can be paired with motion sensors.

- **Contact sensor filter fixed to use `normalize_sensor_type`** — sensor type strings weren't normalized, causing filter mismatches
- **Contact sensor filter updated: only hide those paired with a motion sensor** — standalone door sensors should appear in user app; only those paired as 'window sensor' to a motion sensor should be hidden
- **Cloud backend: hide contact sensors from user app listing** — paired contact sensors are infrastructure (motion sensor helper), not user-facing devices
- **Paired window sensor event leaking to standalone Door Sensor** — when a contact sensor was paired with a motion sensor, its open/close events were also appearing on an unrelated standalone door sensor's log

Note: Resident dropdown removed from contact sensor pairing form.

5. Emergency Button (SONOFF SNZB-01 Zigbee)

One-press SOS alert for the elderly resident.

- **Emergency button alert_log not saving to DB** — button press events were received via Zigbee/MQTT but the alert log entry wasn't being persisted to MongoDB
- **Diagnosed empty alert_logs for emergency button** — traced the full event path from Zigbee2MQTT to MQTT bridge to backend webhook to DB write to locate the break point

6. AltumView Sentinare (Fall Detection Camera)

AI-powered fall detection ONLY. Does NOT do presence, motion, or sleep.

- **"Danger - UNKNOWN" alert fix** — fall detection alerts were showing "Danger - UNKNOWN" instead of proper alert message; fixed alert type mapping on both qa and master branches

Critical correction enforced: AltumView does FALL DETECTION ONLY. Earlier documentation incorrectly attributed presence/motion/sleep capabilities. AltumView provisioning requires WPA2 Personal WiFi.

7. CP Plus Camera (RTSP Video Stream)

IP camera for live video feed via RTSP, proxied through go2rtc on the Pi.

- **Camera stream debugging — COMPLETED** — full end-to-end troubleshooting of RTSP to go2rtc to WebRTC/HLS pipeline on the Pi
- **Camera checker CAMERA_ONLINE/OFFLINE resident lookups fixed** — periodic camera health check was failing to find the associated resident, so online/offline status wasn't reported to backend
- **go2rtc configuration** — set up as part of every Pi deployment for proxying the CP Plus RTSP stream to dashboard via WebRTC

8. Raspberry Pi Hub (Infrastructure)

Central hub running Zigbee2MQTT, MQTT broker, go2rtc, dashboard, and all bridges.

- **Multiple pisetup versions built and delivered:** v26, v27, v27.1, v27.2 (SSD filesystem expansion), v28 (hardware watchdog + network hardening)
- **Fresh Pi setup completed** — full deploy: dashboard + camera + Zigbee + MQTT + GLK + hub heartbeat
- **Home mapping** — POST /api/hub/setup with home_id to bind the Pi to a specific home
- **Zigbee service + cron checkers fixed for home-based device mapping** — checkers were looking up devices by resident instead of by home
- **DS3231 RTC clock chip setup on QA Pi** — hardware real-time clock for accurate timestamps without NTP
- **Hub heartbeat implementation** — periodic heartbeat from Pi to cloud backend so the app knows hub is online
- **SD card corruption root cause analysis** — deep investigation identifying SDIO bus (shared between WiFi and SD card) as hardware-level root cause; formal DOCX report produced
- **SSD migration (USB boot)** — EEPROM bootloader update, Source SATA-to-USB adapter compatibility (UAS quirks, autosuspend disable, blacklist), cmdline.txt + config.txt configuration, live filesystem auto-expansion
- **Hub status log collection** — created hub_status_log MongoDB collection with model + API endpoint for hub health tracking
- **Hardware watchdog (v28)** — enabled Pi's hardware watchdog timer for auto-reboot on hang
- **Network hardening (v28)** — WiFi reconnect scripts, connection monitoring
- **Cudy TR1200 router internet connectivity fix** — diagnosed and fixed QA network internet access
- **Autossh tunnel** — persistent SSH tunnel from Pi to cloud for remote access
- **Mosquitto MQTT broker + Zigbee2MQTT** — configured on each Pi deployment with SONOFF Zigbee dongle

9. Cross-Device / Platform-Wide

Fixes and features spanning multiple devices or the platform as a whole.

- **Resident dropdown removal from ALL device forms except GLK** — simplified pairing flow; home-based mapping replaced per-resident assignment
- **Sync button implementation** — dashboard UI button to force-sync all device states
- **handleSave fix — frontend re-fetches after save** — after saving device config, UI now pulls fresh data instead of showing stale state
- **Device listing ignoring home filter — fixed** — device list API wasn't filtering by selected home

- **"Unauthorized device access" in device_controller.js — fixed** — legitimate device requests were being rejected
- **Alerts route ordering in user_device_route.js — fixed** — route parameter conflicts caused alerts endpoint to 404
- **debug_dashboard_data diagnostic function added** — helper to dump full dashboard data state for troubleshooting
- **Timezone bug in requested_date — fixed** — IST vs UTC offset was shifting which day's data was returned
- **ESLint snake_case enforcement** — applied id-match rule across codebase
- **CI/CD git pull failure diagnosed and fixed** — deployment pipeline was broken
- **EC2 Docker container name conflict resolved** — QA backend container couldn't start due to name collision